

A Double-Switch Single-Transformer Integrated Equalizer for the Recycled Power Battery String of Automatic Guided Vehicles

Bin Xu , Zhaotian Yan, Liang Xiong, Lu Zhang , Wei Zhou , *Member, IEEE*,
Ruikun Mai , *Senior Member, IEEE*, Zhengyou He , *Senior Member, IEEE*,
and Lizhou Liu , *Member, IEEE*

I. INTRODUCTION

Abstract—Applying the recycled power battery to automatic guided vehicles can greatly reduce system cost and prolong the life cycle of the battery. However, the consistency of recycled power batteries is poor, and the imbalance of battery voltage will occur during the charging or discharging process. Therefore, a double-switch single-transformer integrated equalizer is proposed to balance automatic guided vehicles in both charging and discharging operations. The equalizer's multiwinding transformer shares the dc-dc converter's inductor, and the secondary side generates current by utilizing the current ripple on the primary windings. The battery string is balanced by the multiwinding transformer, without extra switches, when the circuit works in charging and discharging mode. A prototype for four battery cells is built to verify the proposed equalizer and its dynamic balance conditions, and a comparison with existing equalizers is presented. The results show that the system in both modes can realize the automatic balance without affecting the function of the dc-dc converter. The proposed circuit requires fewer components than the existing bidirectional integrated equalizer.

Index Terms—Battery management system, cell balancing, current ripple, multiwinding transformer.

THE recycled power battery still keeps 70–80% of the initial capacity [1]. Therefore, the recycled battery is sufficient for the low-speed electric vehicle under the general low charging/discharging current operation conditions [2]. Applying recycled batteries to automatic guided vehicles (AGV) can lower the system cost and prolong the life cycle of the battery [3]–[5]. However, the operating conditions, ambient temperature, and cycle times of the recycled batteries are different, resulting in differences in voltage and capacity [6], [7]. The lower capacity battery is fully charged in the charging mode (CM), while other battery packs are not fully charged, as shown in Fig. 1. When the series-connected battery packs are discharged, the lower-capacity battery pack will first run out of energy. Thus, adding an active equalizer [8], [9] into the AGV can improve the recycled battery's energy utilization.

Existing active balancing methods are divided into two categories [10]: independent equalizer and integrated equalizer. The independent equalizer utilizes energy storage components to deliver energy from the high-voltage battery to the low-voltage battery, resulting in high efficiency. The independent equalizer devices mainly include switched capacitor [11]–[16], multiinductance [17], [18], voltage multiplier [19]–[21], and multiwinding transformer [22]–[29]. These equalizers are composed of independent circuits, which can work in both charging and discharging operations. However, the independent equalizer requires additional switches, components, and controllers, which increases the volume of the energy storage system.

Compared with the independent equalizer, the equalizer integrated into the charging system can simplify the system structure. The integrated equalizer uses the power supply's energy to generate a large current for low-voltage battery and a small current for a low voltage battery to realize the system balance. The integrated equalizer devices include: wireless power transmission (WPT) [30]–[32], voltage multipliers (VM) [33], and multiwinding transformer [34]. Fu *et al.* [30] proposed a multiple-receiver WPT balancing circuit, which achieves the automatic balance for all cells. The multiple-receiver WPT system is advantageous in saving weight and space, ease of implementation, and improved safety. The voltage equalizer based on the WPT and the VM [32] employs capacitors to

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Lu Zhang and Lizhou Liu are with the School of Electronic Engineering, Tsinghua University, Beijing 10086, China (e-mail: willvsnick@tsinghua.edu.cn; liulizhou@my.swjtu.edu.cn).

Bin Xu, Zhaotian Yan, Wei Zhou, Ruikun Mai, and Zhengyou He are with the School of Electrical Engineering, Southwest Jiaotong University, Chengdu 610031, China (e-mail: xubin@my.swjtu.edu.cn; zhaotian_yan@my.swjtu.edu.cn; wzhou@swjtu.edu.cn; mairk@swjtu.edu.cn; hezy@swjtu.edu.cn).

Liang Xiong is with the School of Southwest University of Science and Technology, Mianyang 621010, China (e-mail: xl1989@swust.edu.cn).

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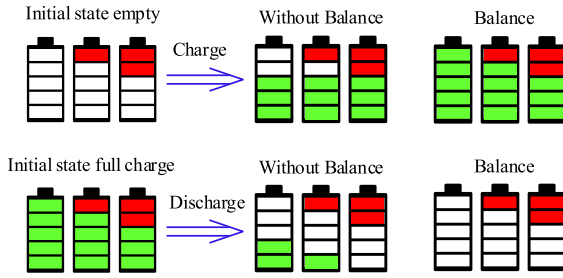


Fig. 1. Example of recycled batterie's voltage status before and after cell balancing.

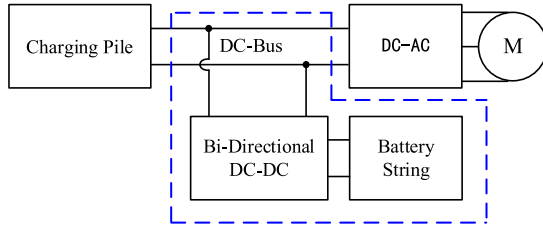


Fig. 2. Circuit structure of AGV.

bridge the cells, improving the balancing speed, and lowering the cost. Uno and Tanaka [33] proposed a graph analysis method of battery balancing systems, which provides theoretical guidance for voltage multiplier compact equalizers.

Hsieh *et al.* [34] proposed a compact-size integrated equalizer with the advantages of high efficiency, compact size, suitable for switching converter, load-related balancing energy, and simple structure without any switch for voltage balance function. A transformer is connected in series at the dc-dc receiving side, and the ripple current generated by the secondary side is used to balance the battery strings. This equalizer can realize a high-current charging for low-voltage batteries. But this equalizer can only realize the balance function in the charging process.

The circuit structure of the AGV is shown in Fig. 2. The charging pile charges the battery string through the bidirectional dc-dc. When the AGV leaves the charging pile, the battery string charges the dc-bus through the bidirectional dc-dc. A tapped inductor (TI) based single magnetic bidirectional PWM converter integrating a voltage equalizer for series-connected SCs is proposed [36]. The TI plays three roles in the proposed integrated converter. The resonant voltage multiplier (RVM) is driven by the square wave voltage generated by the TI. However, the RVM is more suitable for supercapacitor strings as the voltage equalizer because supercapacitor strings have a higher power density.

An equalizer that shares the dc-dc inductor with the balance system by utilizing the current ripple on the primary windings is proposed to minimize the components needed in the proposed system. Additionally, the proposed circuit can balance in both charging and discharging operations, which improves the capacity utilization rate of recycled batteries. When the circuit works in charging and discharging mode, the battery string is balanced

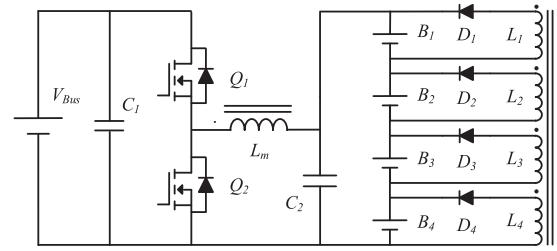


Fig. 3. Structure of the proposed equalizer for four cells connected in series.

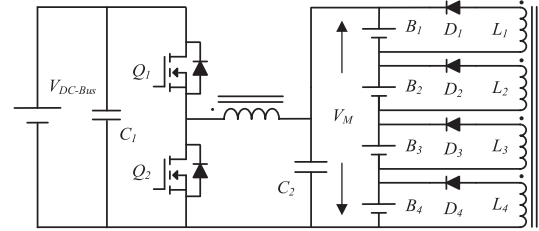


Fig. 4. Working state of the CM of the proposed equalizer.

by the multiwinding transformer. Therefore, the proposed circuit can make full use of the capacity of the recycled battery strings.

II. PROPOSED EQUALIZER

A. Configuration of the Equalizer

As shown in Fig. 3, the proposed equalizer includes two parts: the bidirectional dc-dc converter and the balance circuit. When the charging converter works, the energy can be delivered to the battery through the buck circuit. When the discharging converter works, the battery string supplies the load through the boost circuit. The whole circuit includes two MOSFETs, four diodes, and a multiwinding transformer. The secondary side includes four diodes. The voltage of dc-bus is V_{Bus} , and the current of dc-bus is i_{out} .

- 1) L_m represents the magnetizing inductances of the primary side. L_k represents the leakage inductance of the primary side. L_i , L_{k-i} ($i = 1, 2, 3, 4$), represent the magnetizing inductance and the leakage inductances on a secondary winding, respectively. D_i represents the diodes on a secondary winding. And the transformer can be seen as an ideal transformer.
- 2) The initial cell voltages meet $V_{B,1} > V_{B,2} > V_{B,3} > V_{B,4}$. V_M represents the voltage of the battery string.
- 3) N_1 , N_2 represent the turns of the primary windings and the secondary windings in the transformer. N is the turn ratio of the transformers, expressed as $N = N_1 / N_2$.

B. Charging Mode

When the charging pile charges the AGV, the CM starts. The structure of the proposed equalizer is shown in Fig. 4. In the steady state, the CM has two operating states during one switching period.

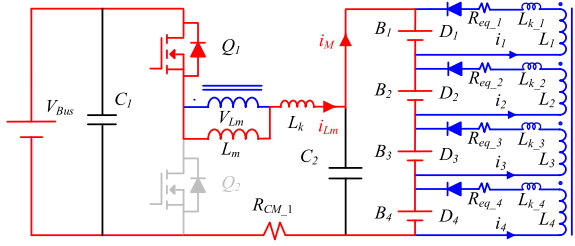


Fig. 5. Operating state of the proposed equalizer in the CM I.

CM I ($t_0 - t_1$): At t_0 , the Q_1 turns ON, and CM I starts.

In this mode, the power supply energy is transferred to the battery string and the inductor L_m through the Q_1 . There is a current ripple in the primary side L_m , which formed the current on the secondary side. The induced current i_i is generated to balance each battery through D_i . Part of the energy is stored in L_m as magnetic energy. The operating state of the proposed equalizer in the CM I is shown in Fig. 5.

The relationship of the V_{Lm} and the V_{L-i} can be expressed as

$$\frac{V_{Lm}}{V_{L-i}} = \frac{N_1}{N_2} = N \quad (1)$$

where V_{Lm} represents the voltage of L_m , V_{L-i} represents the voltage of L_i .

The transformer's secondary windings have the same turn number to achieve equalization among cells in the string. Thus, the voltages of L_i satisfy

$$V_{L-1} = V_{L-2} = V_{L-3} = V_{L-4} = \frac{V_{Lm}}{N}. \quad (2)$$

The voltages of primary winding V_{Lm} can be expressed as

$$V_{Lm} = V_{Bus} - L_k \cdot \frac{di_{Lm}}{dt} - V_M - R_{CM-1} \cdot i_{Lm} \quad (3)$$

where V_M is the battery string's voltage. R_{CM-1} represents the internal resistance of the circuit in the CM I.

V_{L-i} can be expressed as

$$V_{L-i} = V_{B-i} + L_{k-i} \cdot \frac{di_i}{dt} + V_D + R_{eq-i} \cdot i_i \quad (4)$$

where V_D is the voltage drop of diode D_i and i_i is the balance current of B_i . R_{eq-i} represents the internal resistance of the secondary loop

$$R_{eq-i} = R_{L-i} + R_{B-i} + R_{P-i} + R_{D-i} \quad (5)$$

where R_{L-i} is the internal resistance of the inductor L_i , R_{B-i} is the internal resistance of B_i , and R_{P-i} is the circuit bypass loss of the secondary winding. R_{D-i} is the internal resistance of diode D_i .

By solving (4), di_i/dt can be expressed as

$$\frac{di_i}{dt} = \frac{V_{L-i} - V_{B-i} - V_D + R_{eq-i} \cdot i_i}{L_{k-i}}. \quad (6)$$

The secondary balancing current i_i can be derived as

$$i_i(t) = \frac{V_{L-i} - V_{B-i} - V_D}{R_{eq-i}} \left(1 - e^{-\frac{R_{eq-i}}{L_{k-i}}(t-t_0)} \right). \quad (7)$$

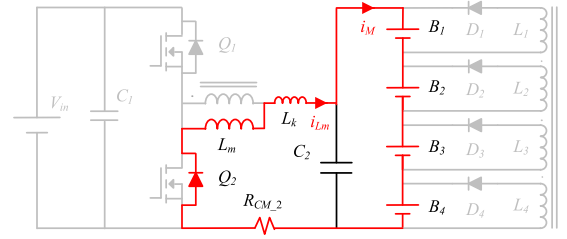


Fig. 6. Operating state of the proposed equalizer in the CM II.

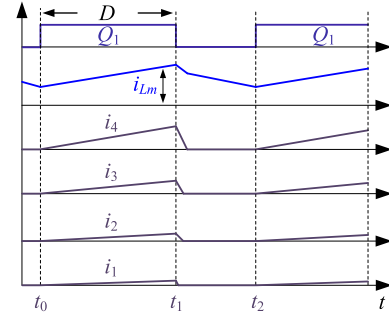


Fig. 7. Key waveforms of the proposed equalizer at CCM in the CM.

Due to the cell's voltage $V_{B-1} > V_{B-2} > V_{B-3} > V_{B-4}$, the balance currents satisfy $i_1 < i_2 < i_3 < i_4$. Therefore, large currents charge the low-voltage batteries, and small currents charge the high-voltage batteries. The equalization for the battery string is automatically achieved by the multiwinding transformer.

CM II ($t_1 - t_2$): At t_1 , both the Q_1 and Q_2 turn OFF, the CM II starts.

The magnetizing energy stored in the inductor L_m is delivered to the battery string through the body diode of the Q_2 . L_m releases the energy, which can automatically realize demagnetization. There is no current through into the secondary winding, and the operating state of the proposed equalizer in the CM II is shown in Fig. 6.

At continuous current mode (CCM), according to the volt-second balance principle, the relationship between voltages in the CM I satisfy

$$L_m \cdot \Delta i_{Lm} = (V_{Bus} - V_M) \cdot DT \quad (8)$$

where D is the duty cycle. Δi_{Lm} is the inductor current increment during the conduction time.

To simplify the analysis, the leakage inductance and V_D are ignored. According to the volt-second balance principle, the relationship between voltages in the CM II satisfy

$$L_m \cdot \Delta i_{Lm} = V_M \cdot (1 - D) T. \quad (9)$$

From (8) and (9), the output voltage of the circuit can be derived as

$$V_M = D \cdot V_{Bus}. \quad (10)$$

The key waveforms of the proposed equalizer at CCM in the CM are shown in Fig. 7.

At discontinuous current mode (DCM), the ON-time of Q_1 in one switching cycle is $D_1 T$. There is current flowing through the

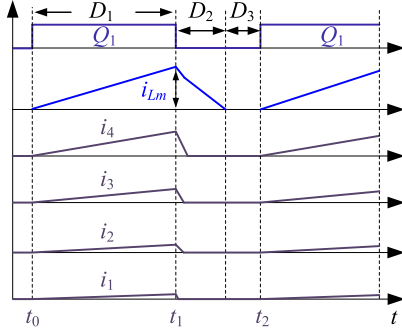


Fig. 8. Key waveforms of the proposed equalizer at DCM in the CM.

inductor L_m at D_2 , and the current flowing through the inductor L_m at D_3 is zero. During the conduction period of the Q_1 , the relationship between voltages in the CM I satisfy

$$L_m \cdot \Delta i_{Lm} = (V_{Bus} - V_M) \cdot D_1 \cdot T. \quad (11)$$

During the period when Q_1 turns OFF, the inductor L_m free-wheels through the body diode of Q_2 , the relationship between voltages in the CM II satisfy

$$L_m \cdot \Delta i_{Lm} = V_M \cdot D_2 \cdot T. \quad (12)$$

From (11) and (12), the output voltage of the circuit can be derived as

$$(V_{Bus} - V_M) \cdot D_1 \cdot T = V_M \cdot D_2 \cdot T. \quad (13)$$

Therefore, the V_M can be expressed as

$$V_M = V_{Bus} \cdot \frac{D_1}{D_1 + D_2}. \quad (14)$$

The charging current I_M is the average value of the inductor current i_{Lm} in one switching cycle, i.e.,

$$I_M = \frac{1}{T} \cdot \left[\frac{1}{2} (D_1 + D_2) T \cdot \frac{V_{Bus} - V_M}{L_m} D_1 \cdot T \right] = \frac{V_M}{R_M} \quad (15)$$

where R_M is the equivalent load of the battery string.

The voltage gain of the DCM dc-dc converter can be obtained as

$$\frac{V_M}{V_{Bus}} = \frac{2}{1 + \sqrt{1 + \frac{8\tau_{L_CM}}{D_1^2}}} \quad (16)$$

where $\tau_{L_CM} = L_m / (R_M T)$.

The key waveforms of the proposed equalizer at DCM in the CM are shown in Fig. 8.

C. Discharge Mode (DM)

When the AGV leaves the charging pile, the battery string charges the dc bus, and the DM starts. The structure of the proposed equalizer is shown in Fig. 9. In the DM, the main circuit is a boost circuit. And the battery string charges the load R_L . Controlling the PWM of Q_2 can adjust the output voltage V_{Bus} . The body diode of Q_1 is used for freewheeling. In the steady state, the DM has two operating states during one switching period.

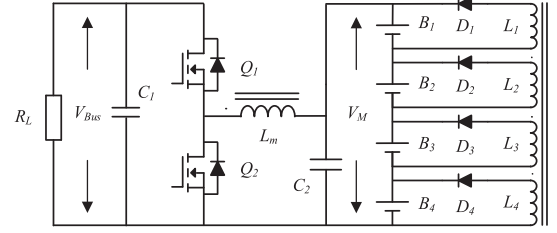


Fig. 9. Structure of the proposed equalizer in the DM.

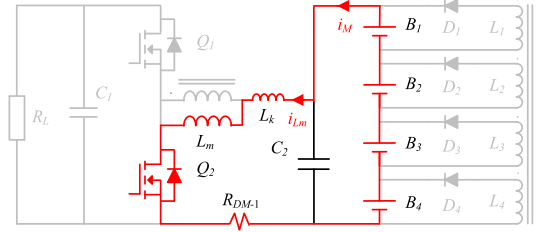


Fig. 10. Operating state of the proposed equalizer in the DM I.

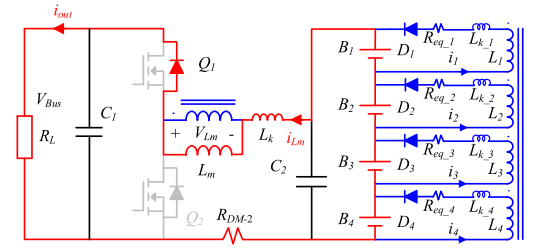


Fig. 11. Operating state of the proposed equalizer in the DM II.

DM I ($t_0 - t_1$): At t_0 , the Q_2 turns ON, the DM I start.

The battery string charges the inductance L_m through the Q_2 . The inductance L_m stores energy in the form of magnetic energy. The operating state of the proposed equalizer in the DM I is shown in Fig. 10.

DM II ($t_1 - t_2$): At t_1 , the Q_2 turns OFF, the DM II starts.

In this state, the battery string and the inductor simultaneously supply load R_L . The equivalent circuit of the DM II is shown in Fig. 11. The energy is transmitted to the load R_L through the freewheeling of the body diode of the Q_1 . The magnetic energy stored in the L_m balances the battery cells through the multiwinding transformer. The induced current i_i is generated to balance each battery through D_i .

The primary winding voltage V_{Lm} can be expressed as

$$V_{Lm} = V_{Bus} - V_M - L_k \cdot \frac{di_{Lm}}{dt} - R_{DM-2} \cdot i_{Lm} \quad (17)$$

where R_{DM-2} represents the internal resistance of the circuit in the DM II.

The relationship between the primary winding voltage V_{Lm} and secondary winding voltage V_{L-i} is determined by

$$\frac{V_{Lm}}{V_{L-i}} = \frac{N_1}{N_2} = N. \quad (18)$$

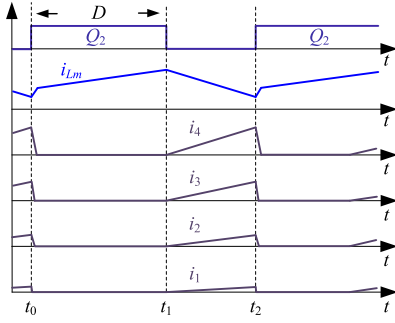


Fig. 12. Key waveforms of the proposed equalizer at CCM in the DM.

Due to the turns number of secondary windings are the same

$$V_{L-1} = V_{L-2} = V_{L-3} = V_{L-4}. \quad (19)$$

V_{L-i} can be expressed as

$$V_{L-i} = V_{B-i} + L_{k-i} \cdot \frac{di_i}{dt} + V_D + R_{eq-i} \cdot i_i. \quad (20)$$

By solving (20), di_i/dt can be expressed as

$$\frac{di_i}{dt} = \frac{V_{L-i} - V_{B-i} - V_D + R_{eq-i} \cdot i_i}{L_{k-i}}. \quad (21)$$

The secondary balancing current i_i can be derived as

$$i_i(t) = \frac{V_{L-i} - V_{B-i} - V_D}{R_{eq-i}} \left(1 - e^{-\frac{R_{eq-i}}{L_{k-i}}(t-t_1)} \right). \quad (22)$$

Due to $V_{B-1} > V_{B-2} > V_{B-3} > V_{B-4}$, $i_1 < i_2 < i_3 < i_4$, which also proves the transformer equalization can achieve balance.

At CCM: To simplify the analysis, the leakage inductance and V_D are ignored. According to the volt-second balance principle, the relationship between voltages in the DM I satisfy

$$L_m \cdot \Delta i_{Lm} = V_M \cdot DT. \quad (23)$$

During the DM II, the primary voltages of the transformer can be roughly expressed as

$$L_m \cdot \Delta i_{Lm} = (V_{Bus} - V_M) \cdot (1 - D)T. \quad (24)$$

From (23) and (24), the output voltage of the bidirectional circuit can be derived as

$$V_{Bus} = \frac{V_M}{(1 - D)}. \quad (25)$$

The key waveforms of the proposed equalizer at CCM in the DM are shown in Fig. 12.

The turn-ON time of Q_2 in one switching cycle is $D_1 T$. There is current flowing through the L_m at D_2 , and the current flowing through the L_m at D_3 is zero. During the conduction period of the Q_2 , the relationship between voltages in the DM I satisfy

$$L_m \cdot \Delta i_{Lm} = V_M \cdot D_1 \cdot T. \quad (26)$$

During the period when Q_2 turns OFF, the body diode of Q_1 turns ON, the relationship between voltages in the DM II satisfy

$$L_m \cdot \Delta i_{Lm} = (V_{Bus} - V_M) D_2 \cdot T. \quad (27)$$

When the inductor current drops to zero, the body diode of Q_1 turns OFF.

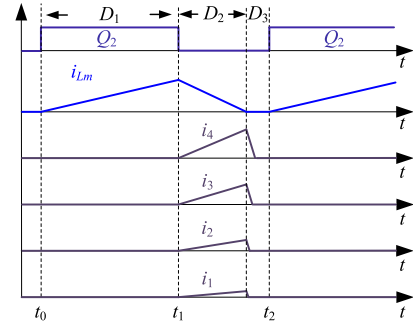


Fig. 13. Key waveforms of the proposed equalizer at DCM in the DM.

From (26) and (27), the output voltage of the circuit can be derived as

$$V_M \cdot D_1 \cdot T = (V_{Bus} - V_M) D_2 \cdot T. \quad (28)$$

The average value of i_{Lm} over the switching period is the average value of the output current I_{out}

$$\begin{aligned} I_{out} &= \frac{1}{2T} \frac{V_M}{L_m} D_1 \cdot T + \frac{1}{2T} \frac{V_M}{L_m} D_2 \cdot T \\ &= \frac{V_M}{2L_m} (D_1 + D_2) D_1 \cdot T \\ &= \frac{(D_1 + D_2) V_{Bus}}{D_2 R_L}. \end{aligned} \quad (29)$$

From (29), the voltage gain of the available DCM boost converter

$$\frac{V_{Bus}}{V_M} = \frac{1 + \sqrt{1 + 2D_1^2/\tau_{L-DM}}}{2} \quad (30)$$

where $\tau_{L-DM} = L_m/(R_L T)$.

The key waveforms of the proposed equalizer at DCM in CM are shown in Fig. 13.

III. DESIGN OF THE SYSTEM

A. Analysis of the Overall Efficiency and Power Losses

Generally, the balance systems' efficiency can be divided into two parts: the CM and the DM. The systems' efficiency can be derived as

$$\begin{aligned} \eta_{CM} &= \frac{V_M I_M + \sum_{i=1}^n I_i \cdot V_{B-i}}{V_{Bus} I_{in}} \\ \eta_{DM} &= \frac{V_{Bus} I_{out} + \sum_{i=1}^n I_i \cdot V_{B-i}}{V_M I_M} \end{aligned} \quad (31)$$

where V_M and I_M are the voltage and current of the battery strings, respectively. I_{in} are the current of the power source. I_{out} is the current of load.

The power losses include transformer loss P_T , conduction loss P_C , and switching loss P_S .

P_C can be divided into bypass resistance loss P_{pass} and diode drop loss P_D . In the CM, the bypass resistance loss P_{pass_CM}

can be expressed as

$$P_{\text{pass_CM}} = \sum_{i=1}^n I_i^2 R_{eq_i} + I_{\text{CM}_1}^2 R_{\text{CM}_1} + I_{\text{CM}_2}^2 R_{\text{CM}_2} \quad (32)$$

where R_{CM_1} , R_{CM_2} are the conduct resistance of the dc-dc circuit in the CM I and CM II, respectively. And I_{CM_1} , I_{CM_2} are the current in the CM I and CM II, respectively.

In the DM, the bypass resistance loss $P_{\text{pass_DM}}$ can be expressed as

$$P_{\text{pass_DM}} = \sum_{i=1}^n I_i^2 R_{eq_i} + I_{\text{DM}_1}^2 R_{\text{DM}_1} + I_{\text{DM}_2}^2 R_{\text{DM}_2} \quad (33)$$

where R_{DM_1} , R_{DM_2} are the conduct resistance of the dc-dc circuit in the DM I and DM II, respectively. And I_{DM_1} , I_{DM_2} are the current in the DM I and DM II, respectively.

The diode voltage drop loss P_D can be expressed as

$$P_D = \sum_{i=1}^n I_i V_D + I_p V_{DQ} \quad (34)$$

where I_p is the current flowing across the body diode of Q_1 or Q_2 . V_{DQ} is the voltage drop of the body diode.

The switching loss P_s consists of two sides, P_R and $P_{\text{ON/OFF}}$. P_R is the conducting losses of the equivalent resistance of the switch. $P_{\text{ON/OFF}}$ is the loss caused by switching ON and OFF of the MOSFET. R_{DS} , E_{OFF} , E_{ON} , U_{DD} , I_D , Q_{RR} , U_{DD} , I_D , and I_{R_D} are some inherent parameters of the MOSFET, which can be found or fitted in the MOSFET's datasheet. P_R and $P_{\text{ON/OFF}}$ can be derived as

$$\begin{cases} P_R = I_{\text{in}}^2 R_{\text{DS}} \\ P_{\text{on/off}} = U_{\text{dc}} \cdot I_{\text{off}} \cdot \left(\frac{E_{\text{on}} + E_{\text{off}}}{U_{\text{DD}} I_D} + \frac{Q_{\text{RR}}}{I_{R_D}} \right) f \end{cases} \quad (35)$$

where I_{in} is the current flowing across the MOSFET. I_{OFF} is the current when the MOSFET turns OFF.

And the core loss $P_{\text{core_loss}}$ can be expressed as

$$P_{\text{core_loss}} = C_m f^\alpha B^\beta C_T \quad (36)$$

where C_m , B , α , β , and C_T are some inherent parameters of the core, which can be found or fitted in the core's datasheet.

B. Designing Turn Ratio

In the CM: V_{L_i} can be expressed as

$$V_{L_i} = V_{B_i} + L_{k_i} \cdot \frac{di_i}{dt} + V_D + R_{eq_i} \cdot i_i \quad (37)$$

To simplify the analysis, ignored internal resistance R_{eq_i} . The secondary balancing current i_i can be derived as

$$i_i(t) = \frac{V_{L_i} - V_{B_i} - V_D}{L_{k_i}} (t - t_0). \quad (38)$$

From (38), the average balancing current I_i in one cycle can be expressed as

$$I_i = \frac{1}{T} \int_0^{DT} \frac{t \cdot (V_{L_i} - V_{B_i} - V_D)}{L_{k_i}} dt$$

$$= \frac{(V_{L_i} - V_{B_i} - V_D)}{2f_{\text{CM}} \cdot L_{k_i}} D^2. \quad (39)$$

From (39), the turn ratio N can be expressed as

$$N = \frac{V_{Lm}}{\frac{2f_{\text{CM}} \cdot L_{k_i} \cdot I_i}{D^2} + V_{B_i} + V_D}. \quad (40)$$

From (40), the smaller the N , the larger the balance current.

The input current of the battery cell is $I_{\text{in}_i} = I_M + I_i$. It is necessary to limit the maximum balance current to exceed the threshold current. Therefore, the minimum value of the N can be expressed as

$$N \geq \frac{V_{Lm}}{\frac{2f_{\text{CM}} \cdot L_{k_i} \cdot I_{i_max}}{D^2} + V_{B_i} + V_D} \quad (41)$$

where I_{i_max} is the maximum balance current of battery.

To ensure that V_{B_i} is lower than $V_{B_i_max}$ after the equalization circuit is fully charged. The value of the N can be expressed as

$$N \geq \frac{V_{Lm}}{V_{B_i_max} + V_D} \quad (42)$$

From (41) and (42), the winding ratio N can be set.

In the DM: V_{L_i} can be expressed as

$$V_{L_i} = V_{B_i} + L_{k_i} \cdot \frac{di_i}{dt} + V_D + R_{eq_i} \cdot i_i. \quad (43)$$

To simplify the analysis, ignored internal resistance R_{eq_i} . The secondary balancing current i_i can be derived as

$$i_i(t) = \frac{V_{L_i} - V_{B_i} - V_D}{L_{k_i}} (t - t_1). \quad (44)$$

From (44), the average balancing current I_i in one cycle can be expressed as

$$\begin{aligned} I_i &= \frac{1}{T} \int_{\text{DT}}^T \frac{t \cdot (V_{L_i} - V_{B_i} - V_D)}{L_{k_i}} dt \\ &= \frac{(V_{L_i} - V_{B_i} - V_D)}{2f_{\text{DM}} \cdot L_{k_i}} (1 - D^2). \end{aligned} \quad (45)$$

In the DM, since the battery is in the discharged state, the balance current is not limited. Therefore, it can increase the balance current by setting the operating frequency of the circuit.

C. Design of the L_m

In the CM: The proposed circuit working in CCM. Under critical conditions, the average current i_{av_CM} flowing through the inductor L_m is the average of the area of the triangle, and current ripple Δi_{L_CM} , which can be expressed as

$$\Delta i_{L_CM} \leq 2i_{av_CM}. \quad (46)$$

When bidirectional dc-dc circuit works in CCM, the inductor current ripple Δi_{L_CM} can be expressed as

$$\begin{aligned} \Delta i_{L_CM} &= \frac{(V_{\text{Bus}} - V_M) I_{\text{CM}_1}}{L_m} = \frac{V_M \cdot I_{\text{CM}_2}}{L_m} \\ &= \frac{(V_{\text{Bus}} - V_M) DT}{L_m} = \frac{V_M (1 - D) T}{L_m} \end{aligned}$$

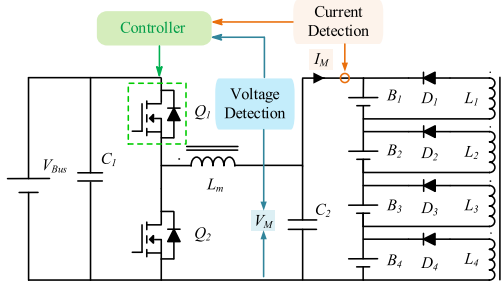


Fig. 14. Control block of the proposed method in the CM.

$$= \frac{V_{Bus} D (1 - D)}{f_{CM} \cdot L_m}. \quad (47)$$

From (46) and (47), Δi_{L_CM} can be expressed as

$$\Delta i_{L_CM} = \frac{V_{Bus} D (1 - D)}{f_{CM} \cdot L_m} \leq 2i_{av_CM}. \quad (48)$$

From (48), the minimum inductance value can be expressed as

$$\frac{V_{Bus} D (1 - D)}{f_{CM} \cdot 2I_{max}} \leq L_m \quad (49)$$

where I_{max} is the maximum current in L_m .

In the DM: The proposed circuit working in CCM. Under critical conditions, the average current i_{av_DM} flowing through the inductor L_m is the average of the area of the triangle, and current ripple Δi_{L_DM} , which can be expressed as

$$\Delta i_{L_DM} \leq 2i_{av_DM}. \quad (50)$$

When bidirectional dc–dc circuit works in CCM, the inductor current ripple Δi_{L_DM} can be expressed as

$$\begin{aligned} \Delta i_{L_DM} &= \frac{V_M I_{DM_1}}{L_m} = \frac{(V_{Bus} - V_M) \cdot I_{DM_2}}{L_m} \\ &= \frac{V_M D T}{L_m} = \frac{(V_{Bus} - V_M) (1 - D) T}{L_m} \\ &= \frac{V_{Bus} D (1 - D)}{f_{DM} \cdot L_m}. \end{aligned} \quad (51)$$

From (50) and (51), Δi_{L_DM} can be expressed as

$$\Delta i_{L_DM} = \frac{V_{Bus} D (1 - D)}{f_{DM} \cdot L_m} \leq 2i_{av_DM}. \quad (52)$$

From (52), the minimum inductance value can be expressed as

$$\frac{V_{Bus} D (1 - D)}{f_{DM} \cdot 2I_{max}} \leq L_m. \quad (53)$$

D. Control of Proposed Circuit

The control of the system is mainly divided into two parts, CM and discharging mode.

In the CM, it is guaranteed that the dc–dc circuit can realize constant voltage and constant current charging battery string. Therefore, the i_M and the V_M are utilized as the feedback signal, which adjusts the duty cycle D of Q_1 , as shown in Fig. 14.

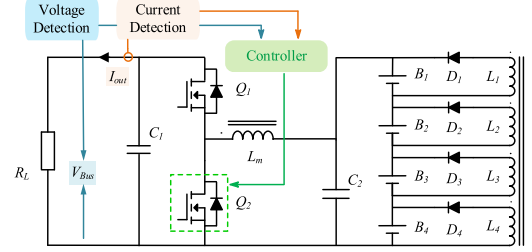


Fig. 15. Control block of the proposed method in the DM.

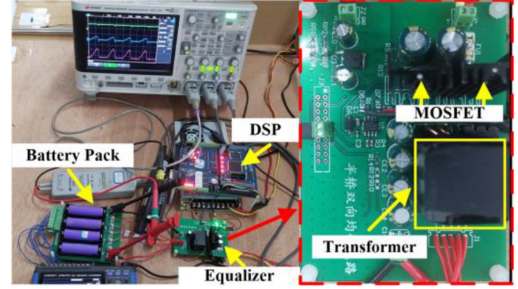


Fig. 16. Photograph of the experimental setup and associated instruments.

TABLE I
SPECIFICATIONS OF THE TRANSFORMER

Symbol	Parameter	Value	Unit
f_{CM}	Charging frequency	180	kHz
f_{DM}	Discharging frequency	120	kHz
V_{Bus}	Input Voltage	26	V
V_B	Battery Voltage	3.2–4.1	V
V_M	The voltage of battery string	12.8–16.4	V
C_1 – C_2	Stabilizing capacitor	940	μF
V_{D1} – V_{D4}	Voltage drop	0.22	V
R_{DS}	Switch resistance	1.1	m Ω
L_m	Primary inductance	25.35	μH
L_k	Leakage of L_m	1.2	μH
L_1 – L_4	Secondary Inductance	4.31	μH
L_{k_1} – L_{k_4}	Leakage of L_i	0.64	μH
N_1 , N_2	5: 2	-	-

In the DM, it is guaranteed that bidirectional dc–dc circuit can realize constant voltage during the discharging process. To ensure that the output power of the battery string is less than the threshold, therefore, the I_{out} and the V_{Bus} are utilized as the feedback signal, which adjusts the duty cycle D of Q_2 , as shown in Fig. 15.

IV. IMPLEMENTATION AND EXPERIMENTAL RESULTS

Fig. 16 shows a photograph of the proposed equalizer applied to four battery cells. IRFB7437PBF MOSFET with 1.1-m Ω drain-source on-resistance is used for switch Q_1 and Q_2 . The controller collects output current and voltage to achieve constant voltage output. The parameters of the balance system are summarized in Table I. The transformer needs the air gap to store magnetic

TABLE II
COMPARISON OF BATTERY EQUALIZERS IN TERMS OF COMPONENTS

Balance methods	Component					
	MOS	Inductor	Capacitor	Diode	Transformer	WPT Coil
Traditional SC [11]	8	0	3	0	0	0
Series-S-capacitor [15]	8	0	3	0	0	0
Time Shared Flyback [29]	7	0	0	7	1	0
Multiwinding Transformer [25]	8	0	0	0	1	0
Global Modular [7]	4	0	0	0	2	0
Unitized Multiwinding [24]	6	0	0	0	1	0
Switch and Transformer [17]	18	0	0	0	1	0
Flyback and Forward [22]	4	0	0	0	2	0
Multiple-Receiver WPT [32]	1	2	9	16	0	1
WPT-VM Based [30]	1	1	8	8	0	1
Single-Switch VM [33]	1	1	5	8	0	0
ICEBC [34]	2	1	0	6	1	0
BCI-RVM [35]	2	0	6	8	1	0
TI bidirectional integrating [36]	2	0	6	8	1	0
Proposed Equalizer	2	0	2	4	1	0

TABLE III
COMPARISON OF BATTERY EQUALIZERS IN TERMS OF BALANCING PERFORMANCES

Balance methods	Efficiency	Balanced Strategy	Working Condition	Cell Current Stress
Traditional SC [11]	92.9%	Automatic	Charge and Discharge	Low
Series-S-capacitor [15]	95.3%	Automatic	Charge and Discharge	Low
Time Shared Flyback [29]	88.0%	Voltage-based	Charge and Discharge	Medium
Multiwinding Transformer [25]	85.3%	Voltage-based	Charge and Discharge	Medium
Global Modular [7]	95.6%	Voltage-based	Charge and Discharge	Medium
Unitized Multiwinding [24]	84.8%	Voltage-based	Charge and Discharge	Medium
Switch and Transformer [17]	80.0%	Voltage-based	Charge and Discharge	Medium
Flyback and Forward [22]	93.0%	Automatic	Charge and Discharge	Medium
Multiple-Receiver WPT [32]	71.0%	Automatic	Charge	Medium
WPT-VM Based [30]	72.8%	Automatic	Charge	High
Single-Switch VM [33]	86.3%	Automatic	Charge	High
ICEBC [34]	92.4%	Automatic	Charge	Medium
BCI-RVM [35]	72.0%	Automatic	Charge and Discharge	High
TI bidirectional integrating [36]	92.1%	Automatic	Charge and Discharge	High
Proposed Equalizer	93.6%	Automatic	Charge and Discharge	Medium

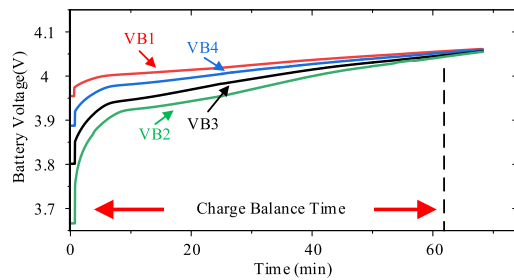


Fig. 17. Experiment verification results of the battery string to four cells in the CM.

energy, and the E-type P95 core is used. A DSP (28335) was used for the digital controller.

The charging experiment verification results of the battery string are shown in Fig. 17. The battery cells' initial voltages are 3.952, 3.662, 3.801, and 3.889 V, respectively. After about 63 min charging balance, the voltages become 4.057, 4.053, 4.054, and 4.057 V. The results show that the voltages gradually tend to be consistent. The voltage difference decreased from 0.29 to 0.04 V. Due to the properties of the transformer, the low-voltage cells will get more energy, and automatically equalized can be

realized. The voltage of the power source is 26 V; the working frequency f is 180 kHz.

Fig. 18 shows the proposed equalizer's experimental waveforms at the switching frequency of $f = 180$ kHz, which agrees with the theoretical analyses. Fig. 18(a) shows the PWM, the charging voltage, the charging current, and i_{Lm} . The current flowing in the inductor L_m gradually increases when the switch Q_1 is connected. The charge current i_M is constant. Fig. 18(b) shows waveforms of balancing current i_1-i_4 . The results show that $i_1 < i_2 < i_3 < i_4$. The low-voltage battery has a large balance current, and the high-voltage battery has a small balance current. The waveforms are consistent with the theoretical analysis of Section II. Therefore, the battery unit's equalization can be realized in the CM. When the switch Q_1 turns OFF, to prevent the balance current from falling, the steepness of i_{Lm} is increased until the balance currents are zero.

The discharging experiment verification results of the battery string to the battery unit scenario are shown in Fig. 19. The initial voltages of the battery cells are 3.821, 3.896, 4.018, and 3.968 V, respectively. After about 50 min charging balance, the voltages become 3.647, 3.649, 3.652, and 3.650 V. The voltage difference decreased from 0.197 to 0.05 V. Due to

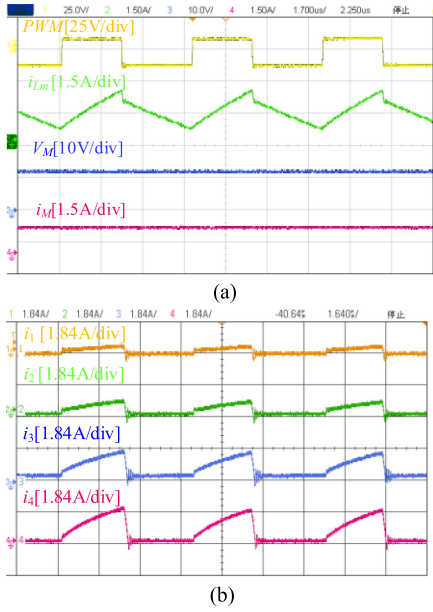


Fig. 18. Forward transform's experimental waveforms of the proposed equalizer at the frequency of $f = 180$ kHz. (a) Waveforms of PWM, charging currents i_{Lm} , V_M , and i_M . (b) Waveforms of balancing current i_1-i_4 .

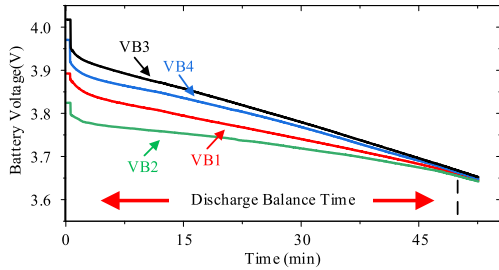


Fig. 19. Experiment verification results of the battery string to four cells in the DM.

the properties of the transformer, the low-voltage battery will get more energy, and the cells can be equalized automatically. The output voltage V_{out} is 26 V; the working frequency is 120 kHz.

Fig. 20 shows the proposed equalizer's experimental waveforms at the switching frequency of $f = 120$ kHz, which agrees with the theoretical analyses. Fig. 20(a) shows the PWM, charging currents i_{out} , output voltage V_{Bus} , and i_{Lm} . Fig. 20(b) shows waveforms of the balance current $i_1 < i_2 < i_3 < i_4$. The results are consistent with the theoretical analysis of Section II. Therefore, the battery unit's equalization can be realized in the discharging mode. When the switch Q_2 turns ON, to prevent the balance current of the secondary winding from falling, the steepness of i_{Lm} is increased until the balance currents are zero.

The CM's efficiency is 94.2%, where the input power is 15.83 W, the output power is 12.07 W, and the balanced power is 2.84 W. While the discharging mode's efficiency is 93.6%, where the input power is 17.24 W, the output power is 11.83 W, and the balanced power is 4.3 W. The calculated power losses of the proposed system are shown in Fig. 21.

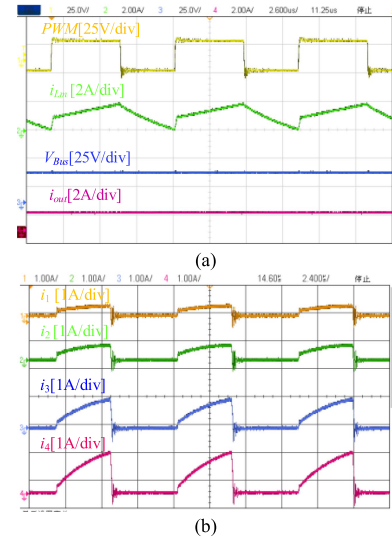


Fig. 20. Transform's experimental waveforms of the proposed equalizer at the frequency of $f = 120$ kHz in the DM. (a) Waveforms of PWM, i_{Lm} , output voltage V_{out} , and i_{out} . (b) Waveforms of balancing current i_1-i_4 .



Fig. 21. Calculated power losses of the CM and DM.

V. COMPARISON WITH CONVENTIONAL BATTERY EQUALIZERS

To identify the benefits of the proposed scheme, Table II compares the proposed equalizer with the traditional solutions regarding the volume of the component. And Table III presents the balancing performances of the proposed equalizer and the traditional solutions, respectively. It is assumed that the battery string has 4 cells. The component numbers and balancing performances of all the equalizers are calculated and evaluated based on the modularization method [29].

In order to make a fair comparison, the size of the equalizers is evaluated at the same balancing power [13]. Therefore, the circuit size mainly depends on the number of components, including MOSFETs, resistors, inductors, capacitors, diodes, transformers, and WPT coil. In Table III, four indexes are employed to evaluate the balancing performances, which are the balancing efficiency, control strategy, working condition, and cell current stress. The control strategy includes voltage-based mode and automatic mode. Voltage-based mode needs to collect the battery cell voltage and control the equalizer to balance. Automatic mode can achieve battery string balance without a single voltage acquisition circuit.

The equalizer relies on the bidirectional buck–boost circuit. However, the disadvantages are that the proposed solution may not be suitable for long battery strings.

VI. CONCLUSION

In this article, an equalizer that shares the dc–dc inductor with the balance system by utilizing the current ripple on the primary windings was proposed to minimize the components needed in the proposed system. Due to the secondary windings having the same turns, the battery string can automatically balance without cell monitoring. In addition, the primary winding releases the magnetic energy into the load, which provides an opportunity for the automatic demagnetization of the forward transformer. The efficiency of the CM and the discharging mode reach 94.2% and 93.6%, respectively.

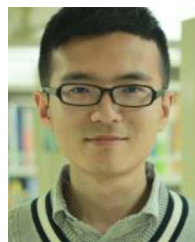
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Bin Xu received the B.S. degree in building electrical and intelligent engineering from the School of Urban Construction, Changchun University of Architecture, Changchun, China, in 2019. He is currently working toward the master's degree in electrical engineering with Southwest Jiaotong University, Chengdu, China.

His main research interests include the battery balancing equalizers, especially on control method and circuit topology.



Wei Zhou (Member, IEEE) received the B.E., M.S., and Ph.D. degrees in electrical engineering from the School of Automation, Chongqing University, Chongqing, China, in 2013 and 2018, respectively.

In 2016, he was a Visiting Scholar with the Department of Electrical and Computer Engineering, University of Auckland, Auckland, New Zealand. He was with the School of Electrical Engineering, Southwest Jiaotong University, Chengdu, China, as a Lecture in 2018. His current research interests include wireless power transfer technology and its applications in rail transit system.

His main research interests include wireless power transfer technology and its applications in rail transit system.



Zhaotian Yan received the B.Sc. degree in electrical engineering and automation in 2018 from the School of Electrical Engineering, Southwest Jiaotong University, Chengdu, China, where he is currently working toward the Ph.D. degree in electrical engineering.

His main research interest includes on wireless power transfer.



Ruikun Mai (Senior Member, IEEE) received the B.Sc. and Ph.D. degrees in electrical engineering from the School of Electrical Engineering, Southwest Jiaotong University, Chengdu, China, in 2004 and 2010, respectively.

He is currently a Professor with the School of Electrical Engineering, Southwest Jiaotong University, Chengdu, China. His research interests include wireless power transfer and its application in railway systems, power system stability, and control.



Liang Xiong received the B.E. and M.S. degrees in control engineering from Southwest University of Science and Technology, Mianyan, China, in 2013 and 2016, respectively.

He is currently a Lecturer with the Southwest University of Science and Technology, Mianyan, China. His main research interest includes energy storage technology.



Zhengyou He (Senior Member, IEEE) received the B.E. and M.S. degrees from Chongqing University, Chongqing, China, in 1992 and 1995, respectively, and the Ph.D. degree from Southwest Jiaotong University, Chengdu, China, in 2001, all in electrical engineering.

Since 2002, he has been a Professor with the School of Electrical Engineering, Southwest Jiaotong University. He was a Visiting Scholar with Cornell University, NY, USA, from 2008 to 2009. His research interests include the area of

signal processing and information theory and its application in electrical power systems, and the application of wavelet transforms in power systems.



Lu Zhang received the B.S. degree in automation and the M.S. and Ph.D. degrees in computer science and engineering from Northwestern Polytechnical University, Xi'an, Shaanxi, China, in 2012, 2015, and 2020, respectively.

From 2016 to 2018, he was a Visiting Graduate Student with the Department of Computer Science and Engineering, University of California, San Diego. He was with the School of Electrical Engineering, Tsinghua University, Beijing, China, as a Postdoc Researcher in 2021. His

research interests include hardware security, design automation, and embedded systems and optimization.



Lizhou Liu (Member, IEEE) received the B.E. degree in electrical engineering and the M.S. degree in control engineering from the Southwest University of Science and Technology, Mianyan, China, in 2014 and 2017, respectively. He is currently working toward the Ph.D. degree in electrical engineering from the School of Electrical Engineering, Southwest Jiaotong University, Chengdu, China.

He was with the School of Electrical Engineering, Tsinghua University, Beijing, China, in 2022.

His main research interests include energy storage technology, especially battery voltage equalizer and hybrid energy storage technology.